Initial Operation and Automation of a Pellet Injection Test Stand

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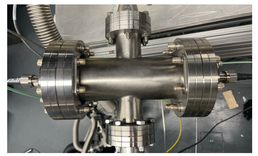
Motivation

- Cryogenic Pellets At Columbia (PAC) test stand has been assembled and operated
- Pellets are injected and ablate into plasma for disruption mitigation, core fueling, density control, and edge-localized mode pacing
- Previous ablation data, primarily in situ from tokamaks, limited by diagnostics and plasma dynamics
- System employs 30 keV electron beam as ablation source
- To-be used with H_2, D_2, Ne, Ar, and mixed pellets
- Modular barrel allows pellet diameter scans
- Redundant mass and mass-loss diagnostics characterize pellet ablation
- Pellet species, geometries, and beam currents will be scanned, producing ablation scaling laws informing models

Diagnostics

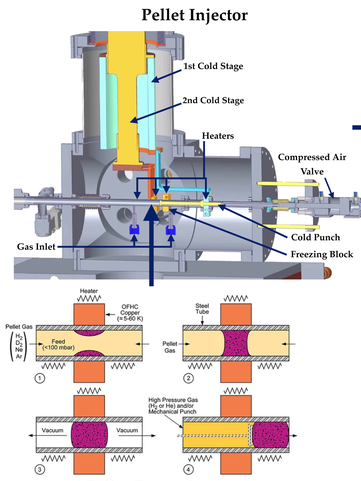
- Baratrons (ideal pressure-difference mass)
- Phantom V1212 fast camera (speed via pixel calibration)
- Tracking frame analysis script (ablation via angle of deflection)
- Resonant microwave cavities (pre and post ablation mass, time-of-flight speed)

Fast Camera

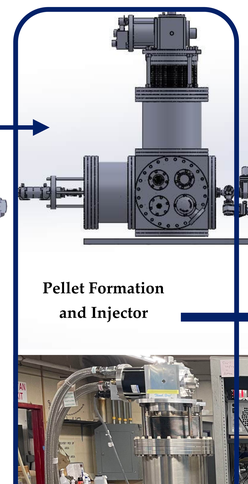


Microwave Cavity

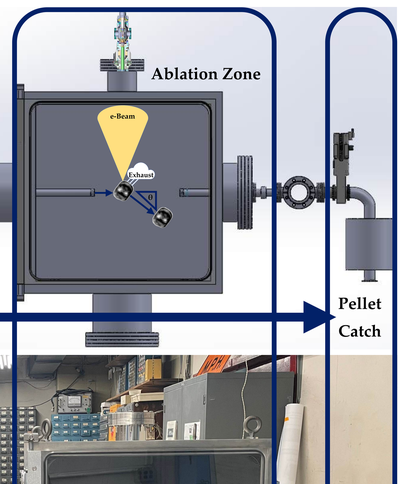
Experimental Design



Pellet Formation [1]

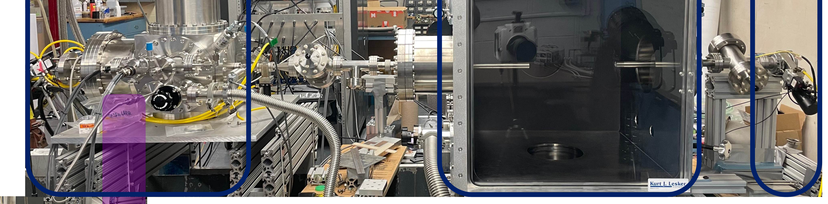


Pellet Formation and Injector

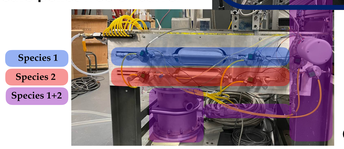


Flight Path

Pellet Catch



- Gas flows from multi-species gas manifold into barrel
- Gas desublimates in concentric monolayers on barrel
- Resistive heaters allow operations near triple point of multiple species
- Actively cooled punch shears pellet free of barrel



Gas Manifold

- Pellet encounters microwave cavity 1, electron beam and fast camera, microwave cavity 2, then pellet catch along flight path
- Rocket effect from top pellet ablation causes deflection down, captured by camera and correlated to ablation
- Pellet catch gate valve actuates closed with punch actuation to mitigate back streaming shards and pressure

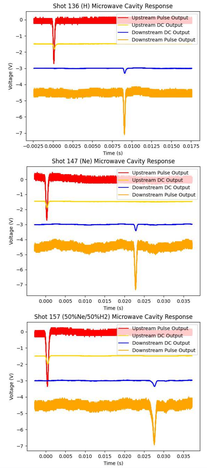
System Characterization

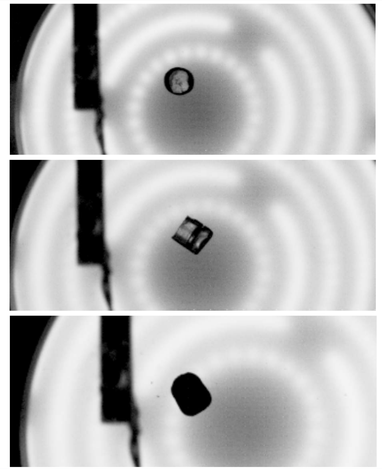
$420\ \mathrm{mm}^3$ Baseline Pellet Scenarios

Species	Ideal Mass (mg)	Average Cavity 1 Mass (mg)	Average Cavity 2 Mass (mg)	T.O.F. Speed (m/s)	Pixel Speed (m/s)	Vertical Dispersion (°)
H_2	41.6 ± 0.2	41 ± 4	41 ± 4	152 ± 11	151 ± 2	-0.2 ± 0.4
Ne	702 ± 2	630 ± 50	580 ± 50	57 ± 4	59 ± 1	-0.4 ± 0.2
50%Ne/50%H_2	330 ± 10	360 ± 60	330 ± 60	48 ± 4	46 ± 1	-0.1 ± 0.4

- 5 pellets of each species formed, equal cryogenic volumes (8.5mm diameter, L/d < 1.0)
- Ideal mass from manifold pressure-volume difference (STP)
- Time-of-flight speed between microwave cavities
- Speed calculated from distance per pixel calibration
- Vertical dispersion from pellet centroid tracking across frames
- Ne cryogenic dielectric constant estimated 1.33 from constant volume calibration

$$\epsilon = \frac{5.57 f_0 V_{\mathrm{pellet}} - 2 V_{\mathrm{cavity}} \Delta f}{V_{\mathrm{cavity}} \Delta f + 5.57 f_0 V_{\mathrm{pellet}}}$$

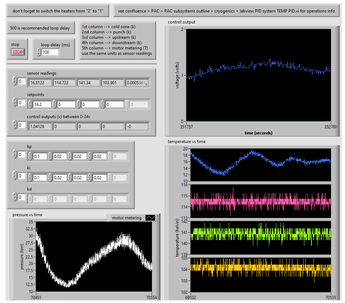


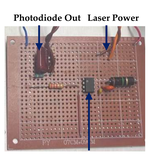


- Initial pellets opaque and brittle from 0.5 mTorr/s leak impurities
- Reducing leak yielded clear, shard-free 1-species pellets
- Mixed pellets opaque, freeze less of fractional ideal mass than 1-species
- Successfully formed H_2 shell Ne core

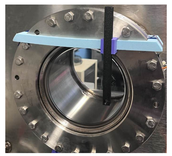
Pellet Formation and Control

- LabVIEW software enables remote control and automation
- Temperature feedback allows size/shape control, required for Ne and Ar injection
- PID gains must be stable for flowing gas heat flux, initial ringing not critical
- Barrel pressure feedback via motor needle valve
- Valve and punch automated actuation referencing manifold and barrel pressures will allow consistent automated formation/injection "recipes"





5V TTL Trigger Pulse



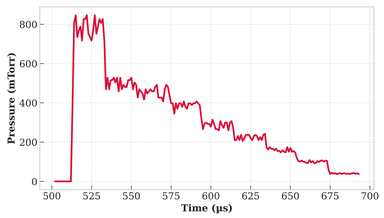
Photodiode Holder

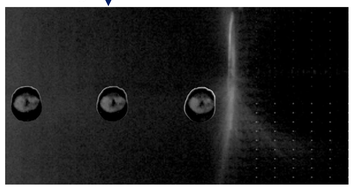
Laser Holder

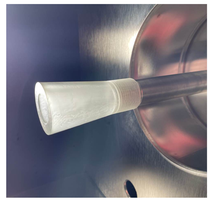
- "Laser tripwire" with photodiode generates voltage pulse when pellet blocks laser
- Circuit and function generator produce adjustable width and delay TTL trigger for e-beam

Ablation Preparation

- Pressure inside beam must be minimal, preventing ion backscattering to e-beam
- Active e-beam pumping and catch systems being explored
- Tullomer and resin printed guide cones prevent reentry shattering







- Determine laser tripwire delay
- Install beam (December) and begin beam current-pellet species scans for scaling laws

[1]: Combs, S. K., and L. R. Baylor. "Pellet-Injector Technology—Brief History and Key Developments in the Last 25 Years." Fusion Science and Technology, 73, no. 4, 2018, 493–518, DOI: 10.1080/15361055.2017.1421367.

This work is supported in part by US DOE under contract DE-AC05-00OR22725 and Columbia University